Mikołaj Sobczyk

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Profile

MSc Computer Security student at VU Amsterdam focused on offensive security: Android app analysis, reverse engineering with Ghidra, and Linux and systems programming.

Education

Vrije Universiteit Amsterdam, MSc in Computer Security *2025 – 2027 (expected)*
- Relevant courses: Binary and Malware Analysis, Software Security, AI for Security

Vrije Universiteit Amsterdam, BSc in Computer Science *2022 – 2025*
- Developed expertise in algorithms, operating systems, and systems programming
- Bachelor's thesis on the dynamic analysis of Google Play Protect (see Projects & Research)

The Australian National University (Exchange Program) *Jul 2024 – Dec 2024*
- Completed courses: Systems, Networks & Concurrency; Cyber Security Foundations; Scientific High Performance Computing; Data Wrangling

Skills

Security: Malware Analysis • Reverse Engineering • Android App Analysis • Software Security • Secure Systems
Programming: Python • C • C++ • Kotlin • Java • Assembly (x86-64, AT&T) • Bash • Scala
Tools & Platforms: Linux • Ghidra • Wireshark • Git • Docker • Android Studio • FastAPI • Google Cloud
Languages: Polish (native) • English (advanced, CAE certificate 2021)

Experience

Junior Technological Solutions Engineer | Super Agency BV (Coliving Ventures) *Jun 2025 – Jan 2026 | Amsterdam*
- Conducted large-scale data collection, processing, and analysis with Python
- Developed automated web scraping pipelines for structured data acquisition
- Explored Retrieval-Augmented Generation (RAG) for AI-driven insights

Teaching Assistant | Vrije Universiteit Amsterdam *Nov 2023 – Feb 2024 | Amsterdam*
- Guided students in programming practices, debugging, and secure coding
- Simplified complex computer science concepts to enhance comprehension

Co-Head of IT | International Congress of Polish Student Societies *Jul 2023 – Feb 2024 | Amsterdam*
- Deployed encrypted ticket management and database system, ensuring data security
- Streamlined IT infrastructure for event operations and cybersecurity compliance

Student in Robotic Process Automation | Nordea *Jul 2023 – Nov 2023 | Łódź*
- Automated manual workflows, reducing human error and improving resilience
- Enhanced operational security through process auditing and monitoring

Projects & Research

Bachelor's Thesis: Bypassing Google Play Protect (grade 7.0, supervisor Prof. Herbert Bos) *2025*
- Built PennyTracker, a benign Android app (Kotlin, Jetpack Compose) that logged every interaction and the device environment to a FastAPI backend, and submitted it to Google Play so the vetting run itself became the dataset
- Found that Google Play Protect's dynamic analysis ran once, for about 6 minutes, on a Google-hosted device, and never exercised form input or deeper app logic
- Concluded that simple environment checks combined with time or logic bombs would likely evade it, and that several weaknesses reported for Google Bouncer in 2012 persist

Hardware Security: Microarchitectural and Embedded Attacks
- Reverse-engineering DRAM bank mappings via row-buffer timing and building cache and speculative side channels
- Extracting and exploiting embedded firmware, group project reproducing a state-of-the-art hardware attack

Binary and Malware Analysis: Reverse Engineering the "Neuralinq" Malware *2026*
- Six-stage reverse-engineering chain: syscall-level binary patching and anti-debugging bypass with HyperDbg, unpacked a packed ELF (ptrace + auxv checks) with a custom Python repair tool
- Built an Intel Pin taint-analysis tool to recover a hidden input, automated control-flow deobfuscation and CFG recovery, and solved a checker with angr symbolic execution

Software Security: Binary Exploitation *2025*
- Wrote working memory-corruption exploits in Python (pwntools) against a series of Linux challenge binaries, from stack buffer overflows to defeating modern mitigations
- Applied return-oriented programming (ROP), GOT/PLT techniques and ASLR bypasses with NX enabled